

MSc(IT) Semester III [Autumn 2024-25]

IT644: Web Services & SOA

Nov 23, 2024

Duration: 2 hours

Max. Marks: 60

Roll Number _____ Name _____

1. Visualize using a UML component diagram, a web app architecture comprising components including API Gateway, Authentication Service, Business Services, and Database Service. Assume that dependencies and interfaces define how these services interact. (3 Marks)

2. Visualize using a UML deployment diagram, a multi-tier architecture for a web app comprising web, application, and data tiers. (3 Marks)

3. Using a UML deployment diagram, depict QAs (availability, redundancy, security) in a web app. (3 Marks)

4. Visualize the publish-and-subscribe design pattern using a UML diagram with an example. (3 marks)

5. Which of the following practices can help manage or reduce technical debt in a software project?
(Select all that apply with reasons) (3 Marks)
- (A) Regular code refactoring.
 - (B) Ignoring non-critical issues to save time by skipping code reviews to accelerate development.
 - (C) Writing automated tests for critical parts of the system.
 - (D) Using continuous integration and deployment.
 - (E) Documenting technical debt and its rationale.

6. In the context of using the **Player-Role Design Pattern** in web applications, which of the following are the correct ways to implement this pattern effectively? (Select all that apply with reasons) (3 Marks)
- (A) Implement roles as independent classes that are composed into player objects dynamically at runtime.
 - (B) Hardcode player-role relationships using conditionals in the player class.
 - (C) Use interfaces or abstract classes to define role behaviors and allow players to adopt multiple roles through polymorphism.
 - (D) Design roles with direct references to the player class for tight coupling to streamline role-specific operations.
 - (E) Separate concerns by implementing a RoleManager that dynamically assigns and manages roles based on runtime conditions.

7. What are the primary advantages of using microservices over a monolithic model? (Select all that apply with reasons) (3 Marks)
- (A) Improved scalability and resource optimization.
 - (B) Simpler deployment and reduced management overhead.
 - (C) Better fault isolation and system resilience.
 - (D) Reduced complexity in application architecture.

8. What are common consequences of technical debt if not addressed in a software project? (Select all that apply with reasons) (3 Marks)
- (A) Reduced software performance.
 - (B) Increased code readability with faster feature delivery in the long term.
 - (C) Higher cost of future development.
 - (D) Enhanced team productivity.
 - (E) Increased risk of defects and bugs.

9. In the context of implementing a singleton design pattern for REST APIs, which of the following are correct practices to ensure effectiveness and maintainability? (Select all that apply with reasons) (3 Marks)
- (A) Use a private constructor and provide a public static method to access the singleton instance.
 - (B) Implement lazy initialization to create the instance only when it is first accessed.
 - (C) Use synchronization or thread-safe mechanisms to handle multiple threads attempting to access the singleton instance.
 - (D) Create multiple instances of the singleton class to support horizontal scaling.
 - (E) Store configuration or shared resources in the singleton to avoid redundancy across API calls.

10. Which of the following statements about Go syntax is correct? (Select all that apply with reasons) (3 marks)
- (A) Go allows implicit type conversion between different types.
 - (B) The `:=` operator can only be used within functions.
 - (C) A variable declared but not used in Go will result in a compilation error.
 - (D) Go supports function overloading.
 - (E) A *defer* statement executes its associated function just before the current function returns.

11. Which of the following aspects are evaluated during portability testing of a software application?
(Select all that apply with reasons) (3 marks)
- (A) Installability
 - (B) Adaptability
 - (C) Maintainability
 - (D) Replaceability
 - (E) Performance Efficiency

12. What is the utility of CIAM systems? Summarize features of three widely used CIAM platforms. (3 marks)

13. How do web and application servers differ?

(3 marks)

14. Which of the following features are supported by the Golang programming language? (Select all that apply with reasons) (3 Marks)
- (A) Garbage collection for automatic memory management.
 - (B) Generics for type-safe programming.
 - (C) Built-in support for concurrency through goroutines and channels.
 - (D) Default function parameters (e.g., allowing functions to have optional arguments).
 - (E) Support for multiple return values from a function.

15. Summarize at least six best practices for implementing RESTful web services. (3 Marks)

16. Which of the following statements best describes the interrelation between different software architecture types? (Select all that apply with reasons) (3 marks)
- (A) Reference architecture can guide the development of product-line architecture by providing a standard framework.
 - (B) Conceptual architecture focuses exclusively on the database design within data architecture.
 - (C) Logical architecture can serve as a precursor to designing physical and deployment architectures.
 - (D) Data architecture plays a crucial role in supporting logical architecture by ensuring that data structures align with functional requirements.
 - (E) Product-line architecture integrates conceptual architecture to manage specific system constraints within a family of products.

17. Define the Usability quality attribute with an example. How do we evaluate usability? (3 marks)

18. Which of the following are critical steps when migrating a monolithic web app to a microservices model? (Select all that apply with reasons) (3 Marks)
- (A) Identify and modularize business capabilities.
 - (B) Adopt a big-bang approach to migration.
 - (C) Implement APIs for communication between services.
 - (D) Use a database-per-service architecture.
 - (E) Ensure centralized logging and monitoring.

19. Summarize best practices to document NFRs. (3 marks)

20. Which of the following factors are critical to evaluating the scalability of web applications? (Select all that apply with reasons) (3 marks)
- (A) Load balancing mechanisms.
 - (B) Database optimization techniques.
 - (C) User interface design principles.
 - (D) Vertical and horizontal scaling capabilities.
 - (E) Test coverage for unit testing.